

Project topic T: Brain Network Communication Models: Effects of Structural Connectivity Rewiring and Information Cost

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This report presents a research-style project into brain network communication models within the framework of complex systems simulation. Several models are used: non-linear neural mass model, random walk, biased random walk, diffusion Laplacian and shortest path. The analysis is based on MRI-derived data from approximately 100 individuals, including structural connectivity matrices and activity time series. Model performance is evaluated by the ability to reproduce empirical functional dynamics from anatomical connectivity. To assess the role of model information cost, the simulation results of empirical structural connectivity is compared with an idealized, shortest-path-optimized rewired network. We evaluate whether rewiring the structural connectivity alters model performance in a manner consistent with each model's theoretical information cost. The results demonstrate that models with low information cost are largely insensitive to structural rewiring, whereas models with medium to high information cost exhibit performance changes of $\Delta r = 0.0215, 0.0646$ and -0.0057 with $p \ll 0.005$. These findings indicate that the impact of structural connectivity rewiring is proportional to the informational demands of the communication model, highlighting the importance of the information cost in structure-function modelling of brain networks.

Project Topic: T - Brain Network Modelling

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I. INTRODUCTION

Neuroscience remains a field characterized by numerous open questions, as many aspects of brain function are still not fully understood [1]. Nevertheless, certain properties of neural systems can be quantified, measured, and simulated. The topic of this report focuses on one such aspect within the subfield of brain network communication modelling. This subfield aims to model and understand how communication occurs in the brain, as well as how the internal functioning of individual neurons scales to large-scale brain networks.

This area of research has a wide range of important and practical applications. In addition to advancing our understanding of human thought, intelligence, and behaviour, it offers valuable insights for fields such as medical diagnosis, sociology, psychology, and artificial intelligence [2]. Recent studies have provided insights into large-scale brain communication phenomena associated with conditions such as depression [3] and epilepsy [1], as well as neurodegenerative and neuro-pathological diseases including Alzheimer's disease, Parkinson's disease, and schizophrenia [1].

In this project, the specific area of brain modelling under investigation is resting-state brain activity and its relationship to the underlying physical structure of the brain. The data used from [4] consist of 100 participants that were scanned under resting state while both their brain activity and physical brain structure were recorded. Brain activity was measured using functional magnetic resonance imaging (fMRI) over a duration of approximately 1.5 hours, capturing Blood Oxygenation Level Dependent (BOLD) signals across 246 Neuron Cluster Regions (NCRs) [4]. In addition, the physical connec-

tions between these NCRs, referred to as Structural Connectivity (SC), were recorded.

Structural Connectivity was obtained using Diffusion Tensor Imaging (DTI), which measures the density of white matter pathways (nerve fibres or axons) connecting NCRs of grey matter. The resulting SC matrix is shown in Figure 1. Notably, the SC is symmetric, as DTI measures large bundles of axons rather than individual fibres, and thus assumes bidirectional connectivity between regions. While this assumption is not strictly accurate, it is considered sufficient given the resolution and limitations of the method [4]. From the fMRI recordings, Functional Connectivity (FC) was computed, representing the correlation of activity between NCRs. An example FC matrix for one participant is also shown in Figure 1. Similar to SC, the FC is symmetric. A partial objective of this work is to model the relationship between SC and FC, where SC serves as the input and FC as the output.

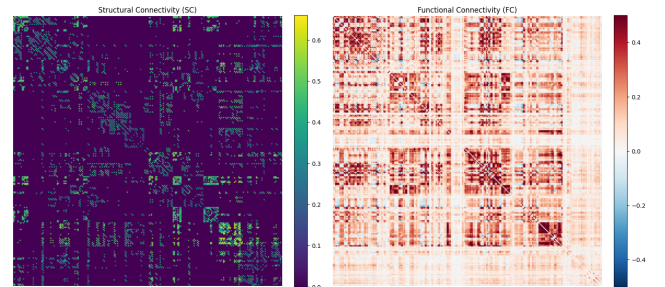


FIG. 1. **Brain network of one person in the data set** provided by [4] Left: Structural connectivity; Right: Functional connectivity (self connections removed for plot clarity)

II. OVERVIEW

The current body of work within this area focuses on identifying models that most accurately reproduce neuronal communication behaviour. Existing approaches, which can broadly be categorised into diffusion-based processes, parametric models, and routing protocol models, have thus far been unable to fully capture these behaviours [5]. Although the classification is somewhat arbitrary, these models—while still grounded in neuronal behaviour—can be evaluated according to three key criteria: delay, energy cost, and information cost [5].

Delay reflects the requirement for rapid communication across the brain network, as high delays introduce inefficiencies. Energy cost represents the metabolic expenditure required to support the chemical processes underlying neuronal signalling, thereby motivating communication strategies that minimise unnecessary activity. Finally, information cost quantifies the amount of global information a neuron would need to possess in order to implement a given model. A high information cost indicates a model that fundamentally fails to reflect the decentralised nature of neuronal behaviour [5]. Based on model assumptions, the information cost is approximately as $RWM < SLM < DLM \approx NLNMM < SPM$.

Model output comparison This method computes functional connectivity from empirical or simulated brain activity and serves as the benchmark against which all models are compared. Model performance is evaluated using the Pearson and Spearman correlation coefficients between the functional connectivity (FC) generated by the models and the FC derived from empirical fMRI data. This evaluation approach is consistent with that used in [1], with the addition of the Spearman coefficient. The analysis primarily focuses on the Pearson correlation, as a direct one-to-one mapping between structural and functional connectivity is expected to exhibit a linear relationship.

Model simulation and optimisation Model parameters are initially optimised using simple optimisation strategies. Two approaches are employed: parameter sweeps and One-at-a-Time Sensitivity Analysis (OATSA). The latter is used for models with too many parameters to feasibly perform a full parameter sweep. parameter sweep optimisation is consistent with [1]. However, for models with more than three parameters, the computational cost becomes prohibitive, as a single simulation can take between 1 and 15 minutes. In such cases, OATSA is used where each parameter is independently perturbed by ± 0.1 around its baseline value, and the resulting FC-FC Pearson correlation is recorded. For all optimisation procedures, the parameter set yielding the highest FC-FC Pearson correlation is selected.

Brain network rewiring and re-simulation The SC matrices of the 100 individuals are subsequently rewired using a Shortest Path Ideal (SPI) algorithm. This algorithm aims to reduce path lengths in Euclidean

space, the physical distances between NCRs, while preserving the fundamental structural characteristics of the brain network. This process requires a careful balance: excessive rewiring would alter the brain structure to such an extent that comparisons with empirical fMRI data become invalid, whereas insufficient rewiring would result in performance changes that are too small to be distinguishable. Following rewiring, all simulations are rerun using the modified SC matrices, and the resulting changes in FC-FC Pearson correlation are analysed for each model.

These three components constitute the core methodology of the project. Figure 2 provides a visual overview of the complete methodological pipeline for illustrative purposes.

III. METHOD

A. Models

As the primary objective of this work is to evaluate differences between models, as many models as possible were included in the analysis. The **SLM**, **NLNMM**, and **DLM** models were implemented programmatically following the descriptions provided in [1], [6], and [7]. In contrast, the **RWM** and **SPM** models were developed during the course of this project based on the methodologies described in [5] and [1]. An overview of all models considered, along with their respective advantages and limitations, is provided in Table I.

The **RWM** is the simplest model and simulates functional connectivity by generating random signals, analogous to a Brownian walk. The signal at each time step is computed as a mixture of its previous value and Gaussian random noise. The model uses two parameters: α , which controls the contribution of the previous signal value, and σ , which defines the variance of the Gaussian noise. This model is equivalent to the SLM with $\beta = 0$.

The **SLM** is an extension of the RWM that incorporates SC through an additional parameter β . By introducing a bias based on the SC, the model transitions from a purely random walk to a biased random walk, placing it within a class of parametric models.

The **DLM**, originally derived in [1], uses structural connectivity to model functional connectivity through an exponential diffusion process. The model contains a single parameter, $\beta\tau$, which jointly represents the diffusion coefficient and the temporal scaling factor. When $\beta\tau = 0$, there is no correlation between NCR activity, whereas when $\beta\tau \rightarrow \infty$, all NCRs exhibit identical activity. Realistic network behaviour is therefore expected for values of $\beta\tau \in (0, \infty)$.

The **NLNMM** is a classical neuronal population model introduced in [6]. It employs 32 parameters derived from biophysical properties of neurons. Each NCR is modelled using three dynamic variables, including the mean membrane potential of excitatory neurons V , inhibitory neurons Z and "open" potassium ion channels

TABLE I. Overview of simulation methods/models.

Method / Model	Use case scenario	Features	Suitable for the project?
Model output comparison	Models output need to be comparable (2 types: activity and FC)	+ FC can be compared for all models - Does not take into account NCR behaviour - Alternative: Using activity + Behaviour is taken into account - Activity is difficult to derive for some model - Models are simplistic and not intended for individual NCR behaviour analysis	FC is a better choice since it is key in Neuronal Communication modelling and is better fit for the models.
Random Walk Model(RWM)	Used to simulate neural NCR activity (simulated BOLD signal) from SC	+ simple + easy to implement + few parameters - poor performance	Representative of the typical diffusion type model. Lowest information cost. Easy to compute.
Stochastic Linear Model (SLM)	Used to simulate neural NCR activity (simulated BOLD signal) from SC	+ simple + easy to implement - poor performance	Improves upon the RWM making it more useful for analysis as its accuracy is better, but increasing its information cost.
Diffusion Laplacian Model(DLM)	Used to simulate FC from SC	+ few parameters + good performance	Provides good performance for a relatively simple model and very different methodology from others.
Non-Linear Neural Mass Model(NLNMM)	Used to simulate neural NCR activity (simulated BOLD signal) from SC	+ good performance + based on neuronal model - many parameters - complex - computationally intensive	The NLNMM has a high interest as it is based on neuronal behaviour, making it highly interesting to include in this project as this is a central part of information cost.
Shortest Path Model(SPM)	Used to simulate FC from SC	+ few parameters + simple	Since the project involves an SPI algorithm, the SPM model is used as another benchmark for how the change impacts the models. The shortest-path model has a high information cost.
SC Shortest Path Ideal (SPI) algorithm	Used to modify empirical SC to test the behaviour on models	+ slightly modifies network + keeps main network structure + terminal NCR are intact - degree is modified - increases the amount of connections	A modification to the SC is needed to test the question, with a reasonable amount of rewiring.

W . making it a 3×246 dimension dynamical system as each NCR is evaluated independently. These variables depend on collective firing rates of excitatory and inhibitory neuronal populations, which are governed by ionic currents associated with calcium, potassium, and sodium electrochemical dynamics. The resulting firing rate of each NCR is transformed into a Blood Oxygenation Level Dependent (BOLD) signal by convolution with a Canonical Hemodynamic Response Function (CHRF). As in [6], each parameter is perturbed by 10% using Gaussian noise independently for each NCR.

The **SPM** computes functional connectivity by weighting the shortest paths of the inverse structural connectivity matrix with an exponential decay function parametrised by α , reflecting the assumption that functional connectivity decreases with increasing network dis-

tance.

B. Optimization

All models include at least one free parameter requiring tuning. This is performed during the optimisation stage. Table II summarises the optimisation methods and parameter ranges used for each model. Parameter sweeps represent the simplest optimisation approach and is used whenever computationally feasible. This method consists of evaluating all combinations within the parameter space defined by $\mathcal{S} = P_1 \times P_2 \times \dots \times P_k = \prod_{i=1}^k P_i$, where $P_i = [a_i, b_i] = \{x \in \mathbb{R} \mid a_i \leq x \leq b_i\}$, discretised using a step size δ_i , and k denotes the number of

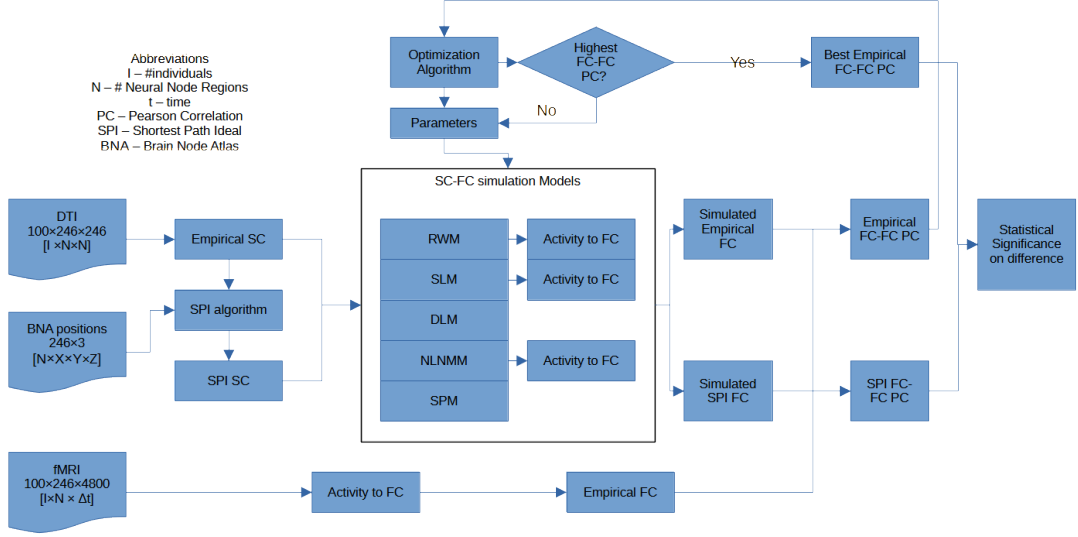


FIG. 2. **Overview of the methodology used.** The DTI and fMRI data is provided by [4]. This is then transformed into SC, SPI SC and empirical FC. The SC and SPI SC are fed to the models and the resulting FC is compared using Pearson correlation coefficient and then t-scored.

Model	Parameters	Method	Values
RWM	α, σ	Sweep	$\mathcal{S} = [0.1, 1] \times [0.1, 1]$ $\delta_\alpha = \delta_\sigma = 0.1$
SLM	α, β, σ	Sweep	$\mathcal{S} = [-3, 3] \times [0, 6] \times [0.1, 2]$ $\delta_\alpha = \delta_\beta = \delta_\sigma = 0.1$
DLM	$\beta\tau$	Sweep	$\mathcal{S} = [0.1, 6]$ $\delta_{\beta\tau} = 0.1$
NLNMM	$\mathcal{P} = P_1, \dots, P_k$ $k = 32$	OATSA	$\mathcal{S} = [0.1, 1.0] \times [0.1, 1.0]$ $\delta_\alpha = \delta_\sigma = 0.1$
SPM	α	Sweep	$\mathcal{S} = [0.1, 2.0]$ $\delta_\alpha = 0.1$

TABLE II. Optimization methods and values for models

parameters. Since the number of required simulations scales as $\mathcal{O}(|P_i|^k)$ with increasing dimensionality, this approach was limited to models with at most three parameters and relatively low computational cost. For models with more than three parameters or high computational demands, the OATSA method was employed. In this approach, simulations were first performed using baseline parameter values obtained from the source literature, after which each parameter was independently perturbed by ± 0.1 . Parameters exhibiting the greatest influence on model performance were subsequently fine-tuned manually.

C. SPI algorithm

The SPI algorithm modifies the empirical SC by rewiring connections. It operates in the following way:

1. Pick a random NCR i . The randomness is to reduce bias. The same NCR cannot be picked twice in a

run.

2. Find all existing paths $\mathbf{C}_{ij}^{(e)}$ from i to j of length n .
3. Find all potential paths $\mathbf{C}_{ij}^{(p)}$ from i to j of length n .
4. Exclude potential paths where the end terminal is changed, $\mathbf{C}_{in}^{(p)} \notin \prod_{j=1}^n [\mathbf{C}_{ij}^{(e)}]$.
5. calculate the Euclidean path length of existing and potential using the BNA Atlas as seen in Figure 2.
6. Rank and sort the paths by distance and rewire the connections of NCR i to the shortest paths, keeping the same values and same amount of connections to that NCR, ensure symmetry.
7. Go to 1 again until all NCRs are processed.

For this implementation, n was chosen to 2, since 3 added a much larger computational load. This algorithm was chosen because, rather than simply rewiring the NCR connections to the closest path, it will keep the destination up to 2 NCRs away intact, ensuring that the network is optimized rather than completely rewired. The amount of change is visually shown in the graph plot in Figure 3.

IV. RESULTS AND DISCUSSION

The SPI algorithm appears to modify the SC network through a mild optimisation process, which aligns with its intended purpose, as more substantial alterations would weaken the biological relevance of the resulting

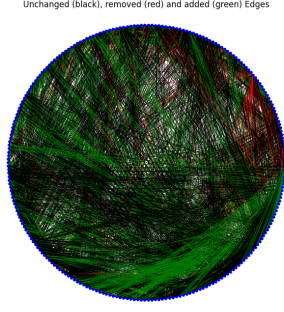


FIG. 3. **Result of the SPI algorithm on one of the individuals** Shows how the SPI algorithm has rewired the SC, green is new connections, red is removed and black is unchanged.

network. For both the original and rewired SC networks, the Mean Path Length (MPL) and Clustering Coefficient (CC) were computed. The desired outcome was minimal change in CC, accompanied by a slight reduction in MPL. This behaviour is observed in Figure 4.

Despite achieving the intended reduction in MPL, the SPI algorithm introduces some less desirable modifications to the network. Specifically, the number of connections increases by 28.40%, which in turn results in a 28.06% increase in the total node degree. Although these changes are not ideal, they may be acceptable as a trade-off, as they arise from the algorithm's preference for preserving end terminals. This constraint necessitates the addition of connections in cases where NCRs exhibit differing ideal shortest paths.

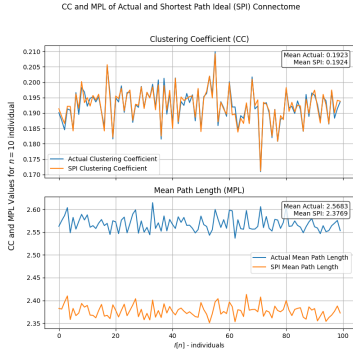


FIG. 4. **Plot of the results from SPI algorithm.** shows the MPL and CC of the SC before and after application of the SPI algorithm for all individuals

The optimisation process yielded substantial improvements across all models, with each showing a significant increase in performance following parameter tuning. The average Pearson correlation coefficients for each model are presented in Figure 5. The observed FC-FC correlation patterns closely resemble those reported in [1],

suggesting that the models have been implemented correctly. Although the Pearson correlation coefficients are generally lower than those reported in [1], this discrepancy may be attributed to differences in the dataset used or to the computation of Pearson correlation using only the upper-triangular portion of the FC matrix, reflecting its symmetry. When the full FC matrix is used, the resulting Pearson coefficients are consistent with those reported in [1].

Following the application of the SPI algorithm, all models were re-simulated using the rewired SC matrices. The resulting changes in model performance are shown in Figure 6 for all models.

To quantitatively assess these changes, a t -test was performed on the difference

$$\mathbf{D} = \mathbf{r}^{(\text{emp})} - \mathbf{r}^{(\text{spi})},$$

where r_i denotes the Pearson correlation coefficient for individual i . The resulting statistics are reported in Table III.

TABLE III. Statistical summary of model results

Model	Mean Difference (mean($\mathbf{D}$))	t -statistic	p -value
SLM	-0.0008	-0.8645	3.89×10^{-1}
DLM	0.0215	18.6037	4.38×10^{-34}
NLNMM	0.0646	9.8411	2.43×10^{-16}
RWM	0.0007	0.7571	4.51×10^{-1}
SPM	-0.0057	-4.2975	4.05×10^{-5}

Table III indicates that the RWM and SLM models exhibit statistically insignificant differences in performance, with mean differences sufficiently close to zero to be reasonably approximated as negligible. In contrast, the DLM and NLNMM models demonstrate a statistically significant decrease in performance following SC rewiring, while the SPM model exhibits a small but statistically significant improvement in performance.

V. CONCLUSIONS AND OUTLOOK

The goal of this project was to investigate the effect of SC rewiring on model performance for the five selected models. The primary focus was on how this effect relates to each model's information cost. The models differ substantially in their theoretical information requirements: the RWM is expected to have the lowest information cost, while the SPM has the highest. The information costs of the DLM and the NLNMM are not explicitly defined; however, given their strong dependence on SC, they are reasonably expected to fall within the medium-to-high information cost range. The SLM, by contrast, is closely related to the RWM but incorporates an SC-based bias, suggesting a low-to-medium information cost.

This relationship is reflected in the results obtained from the Pearson correlation differences following SPI-based

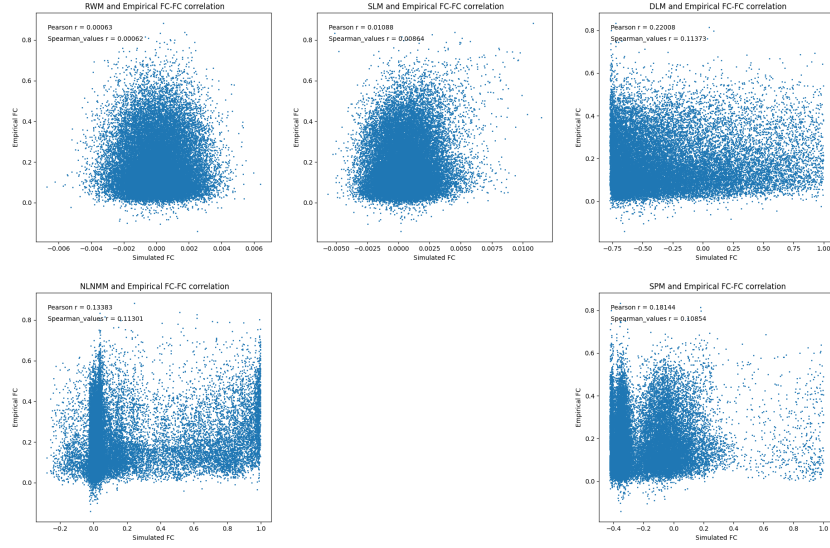


FIG. 5. **FC–FC plots for all individuals.** Should be 1-1 linear as simulated FC of a NCR match actual FC. Includes average Pearson and Spearman correlation coefficients

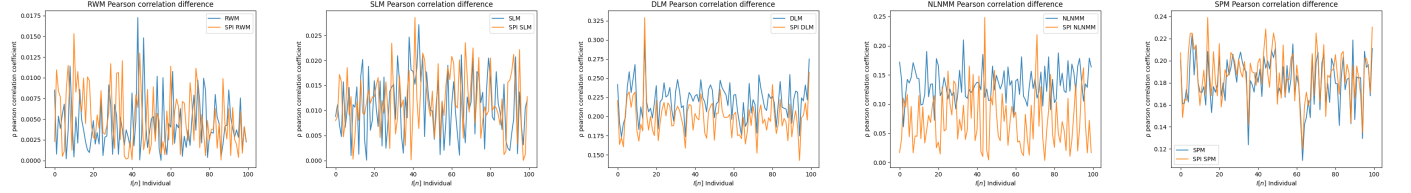


FIG. 6. **Difference in model performance** Difference for each Pearson when simulating on empirical SC compared to rewired SPI SC

SC rewiring. The RWM and SLM exhibit effectively no change in performance (zero difference with minor noise), whereas models with medium-to-high information costs show a significant impact from the SPI algorithm. An exception to this trend is the SPM, which demonstrates a slight improvement in Pearson correlation that exceeds statistical significance ($|t| \geq 2$). While notable, this improvement is not entirely unexpected. The SPM is explicitly based on shortest paths, and the SPI algorithm performs fine-tuning and optimization of network wiring while introducing additional connections. As a result, the rewiring procedure inherently biases the network toward structures that favour shortest-path-based models and possibly routing models in general. Overall, these findings support the central hypothesis that SPI SC rewiring affects model performance in a manner proportional to the model’s information cost, i.e. $\Delta r_{\text{model}} \propto \text{information cost}$. From a mechanistic perspective, this result is intuitive: models that integrate information from a large portion of the network and operate on a global scale are more sensitive to changes in SC, whereas models relying primarily on local or limited information are largely unaffected by modifications to global network structure.

While these results are promising, there remains scope for

further improvement. In particular, the SPI algorithm could be evaluated using multiple variants to compare performance and mitigate the limitations of the current implementation described in Table I. As well as this, assessing the models and adding more models with other properties, for example another routing model that is not shortest path based could counteract the SPM bias.

VI. CONTRIBUTIONS

Johan Maté Conceived the project and implemented the code in Python.

VII. CONFLICT OF INTEREST

The contributors have no known conflicts of interest.

VIII. DATA AND CODE AVAILABILITY

All source code are on: https://github.com/EngyMate/MPCAS_SOCS_Project_BrainNetworkModeling

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